

Technical Supplement for “GAWorld-Bench: A Layered Validation Framework for LLM-Based Artificial Societies”

Anonymous submission

Purpose and Scope

This supplement documents the validation protocol and artifact audit reported in the main paper. It is designed to let a reviewer verify the reported numbers without launching a simulator or calling an LLM. No result in this supplement was generated by a new simulation run. Repository paths are stated relative to the project root in the anonymous artifact package.

The audit distinguishes four evidence-record classes:

- **REAL**: an artifact emitted by an existing GAWorld run;
- **DIAGNOSTIC**: a re-analysis of a real artifact for measurement or failure analysis;
- **SYNTHETIC**: a fabricated structural fixture used only to exercise benchmark code; and
- **INCOMPLETE**: an absent, interrupted, or statistically insufficient artifact.

Evidence is labeled before scoring. Synthetic and incomplete items never enter a model-validity result. Here **DIAGNOSTIC** denotes a declared transformation of a source artifact, not a competing claim about how that source was generated. Thus one real report can support both a real descriptive record and a diagnostic derivative, provided the source-to-statistic relation is explicit.

The audit snapshot was assembled on 2026-07-11 from repository HEAD 424cd46ae5c7, but the working tree contained changes and therefore that commit does not identify the audited bytes. Full SHA-256 values in the package README are the controlling identifiers. We searched the benchmark reports and scorecard, current economy outputs, and known emotion, network, and memory experiment-result directories needed by the five configured tracks. This is a purposive audit of benchmark-relevant evidence, not an exhaustive inventory of every log.

Every scientific test returns one of {pass, fail, unassessed}. Separately, the reproducibility gate returns {OK, UNVERIFIED, UNTRUSTWORTHY}. We therefore report (outcome, gate, coverage) rather than hybrid labels such as “fail/unverified.”

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Formal Validation Protocol

Track A: Macro Empirical Fit

For metric i , let x_i be the simulated aggregate, a_i an external anchor, and $\tau_i > 0$ a relative tolerance. The implemented bounded score is

$$s_i = \max \left(0, 1 - \frac{|x_i - a_i|}{|a_i| \tau_i} \right). \quad (1)$$

The track diagnostic is the arithmetic mean of the observed s_i , subject to configured hard invariants such as money conservation. A result must additionally state:

1. the number of underlying agent observations;
2. the definition and source of each anchor;
3. whether the target was directly parameterized; and
4. which metrics were configured but unavailable.

An input-derived match is classified as a calibration or implementation check. It is not evidence that an unencoded group-level pattern emerges.

The current harness configures an urban Engel coefficient anchor of 0.288 and a savings-rate anchor of 0.35. The savings definition is explicitly described as measurement-sensitive in the benchmark design; it must not be treated as a universal ground truth.

Track B: Emergent Dynamics

Track B is specified but not implemented in the audited version. A valid stylized-fact test requires a predeclared statistic g , horizon T , tolerance or hypothesis test H , and minimum number of seeds R :

$$\mathcal{T}_B = (g, T, H, R, \text{data source}). \quad (2)$$

Candidate tests include logistic diffusion fit, network modularity and homophily slopes, and distributional tail shape. A report based only on initialization, co-location, or a missing state file is incomplete.

Track C: Counterfactual Validity

For an event run E and matched baseline B , the implemented endpoint contrast is

$$\Delta_{e,m}^{\text{final}} = y_{m,T}^E - y_{m,T}^B. \quad (3)$$

The archived comparison files also store the whole-run contrast

$$\Delta_{e,m}^{\text{whole}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} (y_{m,t}^E - y_{m,t}^B). \quad (4)$$

Neither estimand is universally preferable. The endpoint is vulnerable to terminal noise, while the whole-run mean can dilute a late intervention with pre-event observations. An evaluation must state which estimand implements the scientific question.

For preregistered direction $d_{e,m} \in \{-1, +1\}$, a single-run sign check is

$$I_{e,m} = \mathbf{1}[d_{e,m} \Delta_{e,m} > 0]. \quad (5)$$

In multi-seed mode, the item is supported only when the interval estimate excludes zero and the mean has the expected sign. With one observation, the item is statistically unassessed even if its point estimate has the expected direction.

A placebo event uses threshold ϵ and passes metric m if $|\Delta_{e,m}| < \epsilon$. The current implementation defaults to $\epsilon = 0.05$. Domain-specific work should justify or replace this value. The audited protocol does not yet preregister a confidence level, interval estimator, minimum seed count, multiple-comparison rule, or scale-aware equivalence margin. Until those choices are fixed, the multi-seed path is a software capability rather than a completed inferential protocol.

In the currently implemented deterministic mode, same-seed comparison indexes state rows by (agent, step, metric) and reports the fraction of intersecting rows equal within absolute tolerance. No intersecting rows is failure, not success. This subtest is not a universal requirement for stochastic simulators: a stochastic mode should preregister a replicated distributional agreement test instead. Relative to the declared mode, the gate state is:

OK if determinism is assessed and passes, (6)

UNVERIFIED if determinism is not assessed, (7)

UNTRUSTWORTHY if determinism is assessed and fails. (8)

Let $c = n_{\text{eval}}/n_{\text{configured}}$. Coverage is reported beside the component result and is not multiplied into it, because multiplication would be equivalent to imputing every unassessed item as zero. Coverage cannot repair a failed declared trust protocol.

Track D: Individual and Memory Consistency

The proposed probes are profile agreement, traceable recall, contradiction rate, and behavior under memory ablation. For claim q about an agent’s remembered event, traceability is

$$\text{trace}(q) = \mathbf{1}[\text{a timestamped episode supports } q]. \quad (9)$$

If an LLM judge scores coherence or persona agreement, the judge prompt, model version, order effects, and calibration against human labels must be reported. The audited harness does not yet implement this track.

Track E: Reproducibility and Cost

Track E includes cross-seed coefficient of variation, wall-clock and token cost per agent-day, structured parse failure rate, and a mode-appropriate reproducibility test. The audited version implements only the deterministic-mode subtest through Track C. Therefore, Track E is incomplete.

Expanded GAWorld Audit

Macro Snapshots

The current `wealth_snapshot.csv` contains one row:

Artifact	n	Engel	Savings
Current snapshot	1	0.48	0.05
June 15 report	1	0.30	0.25
Synthetic fixture	5	0.29 mean	0.328 mean

The first two rows are real artifacts from different repository states. The third is generated by `make_synthetic` and is excluded from empirical scoring. The current conservation audit contains one row with absolute drift 0.0. The 0.288 Engel target is linked to an external statistic; the 0.35 savings target is an internal, definition-sensitive benchmark setting without a matched external source.

Counterfactual Reports

The archive does not preserve a common comparability key across these dates: code revision, prompt/provider, and population are unknown for all three reports, while baseline lineage is only partial. Accordingly, differences across dates are treated as non-comparable archived observations, not estimates of random-seed instability.

The June 20 report repeats the June 15 traffic and tax values; their lineage is not recorded, so the table does not count them as independent observations. It labels its four directions as passing, but its gate is explicitly `UNVERIFIED`: no placebo or determinism pair was supplied, and several interrupted directories are listed. The July 9 report enters the multi-seed reporting path with $n = 1$ for each item. It correctly gives zero significance coverage. Treating that report as a four-seed or four-intervention confidence panel would be an error because each intervention-metric cell has only one observation.

Synthetic Fixture Fingerprint

The single-run fixture writes these exact contrasts:

Intervention / metric	Synthetic Δ
Traffic / mobility intent	+0.08
Layoff / economic security	−0.12
Layoff / stress	+0.09
Tax / economic security	+0.06

It also writes four placebo contrasts below 0.05 and two identical four-row baseline state files. The current headline scorecard reproduces these values and reports a perfect placebo and determinism score. This exact fingerprint is strong evidence of fixture-like output, but it does not establish lineage without a generation log or input manifest. It does not imply

Table 1: Artifacts used by the audit. “Allowed claim” is deliberately narrower than the content of some legacy reports.

ID	Class	Source	Allowed claim
E-A1	Real	wealth_snapshot.csv	Current one-agent macro snapshot; no distributional inference
E-A2	Real	conservation_audit.csv	Zero drift for one recorded audit row only
E-A3	Real	report_20260615T005429.md	Earlier one-agent macro values differ from the current snapshot
E-C1	Diagnostic	Same June 15 report	Whole-run and endpoint estimands can differ in magnitude and sign
E-C2	Real	report_20260620T133943.md	Point estimates from one historical run per intervention
E-C3	Real	report_20260709T075428.md	Later one-observation batch is incompatible with statistical inference
E-S1	Synthetic	Single-run fixture	Single-seed harness path and schema execute
E-S2	Synthetic	make_synthetic_multiseed	Multi-seed statistical path accepts structured fixtures
E-B1	Incomplete	Emotion comparison JSON	Track B is unassessed because required state files are absent
E-D1	Incomplete	Memory comparison report	Track D is unassessed because one treatment completed both phases

Table 2: Selected real report values. Each item is one observation; the dates identify archived reports, not replicated experimental panels.

Report	Intervention / metric	Whole	Final	n	Direction	Status
2026-06-15	Traffic / mobility intent	+0.0068	+0.3368	1	expected +	Window-sensitive
2026-06-15	Layoff / economic security	+0.0196	+0.1330	1	expected −	Opposite
2026-06-15	Layoff / stress	−0.0329	−0.1920	1	expected +	Opposite
2026-06-15	Tax / economic security	−0.0086	+0.0201	1	expected +	Sign changes by window
2026-06-20	Layoff / economic security	−0.0319	−0.0498	1	expected −	Point estimate only
2026-06-20	Layoff / stress	+0.1446	+0.1717	1	expected +	Point estimate only
2026-07-09	Traffic / mobility intent	—	+0.0000	1	expected +	Unassessed
2026-07-09	Layoff / economic security	—	+0.0826	1	expected −	Unassessed
2026-07-09	Layoff / stress	—	+0.0078	1	expected +	Unassessed
2026-07-09	Tax / economic security	—	+0.0279	1	expected +	Unassessed

tampering or a benchmark bug: the fixture is documented as a no-LLM end-to-end verification mode. The interpretive risk arises only if the scorecard is treated as empirical model performance.

The multi-seed fixture separately defines five values for each intervention-metric pair. Three groups are tight and have the expected sign; the tax group straddles zero to exercise the non-significant code path. Those values are likewise excluded from cross-seed robustness claims.

Offline Verification Procedure

The following checks read existing files and do not start GAWorld:

```
cd /path/to/anonymous/GAWorld
r=benchmark/results/reports
r=$r/report_20260615T005429.md
rg -n '0\.3368|0\.0068' "$r"
rg -n '0\.0201|−0\.0086' "$r"
rg -n '0\.08|−0\.12|0\.09|0\.06' \
  benchmark/gaworld_bench.py \
  benchmark/results/scorecard.json
```

The current macro row can be inspected with Python’s standard library:

```
import csv
from pathlib import Path
p = Path('output/economy')
p /= 'wealth_snapshot.csv'
rows = list(csv.DictReader(p.open()))
print(len(rows))
print(rows[0]['engel_coefficient'])
print(rows[0]['savings_rate'])
```

Expected output is 1 0.48 0.05. The conservation audit analogously contains one row with drift 0.0.

Live benchmark commands that include `−run` require an LLM provider and are outside this audit. No such command was executed to prepare the paper.